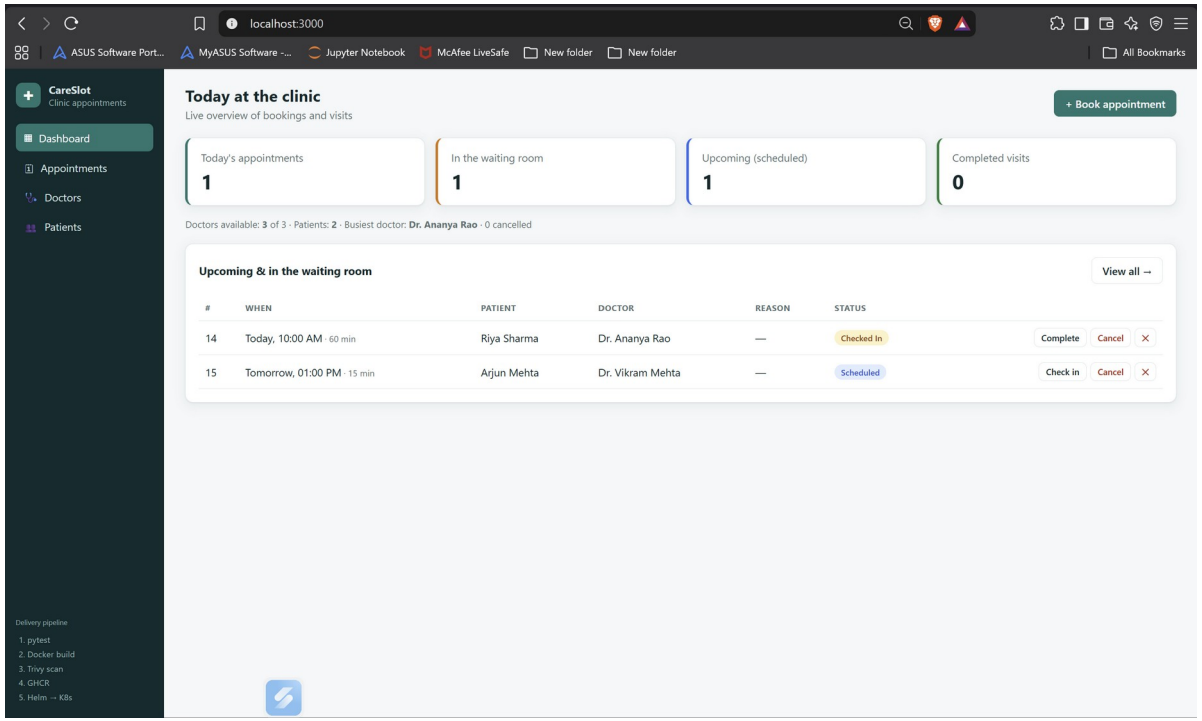
CareSlot

Clinic appointment system - DevOps final project

Palak Agrawal · 24BCS10504 · Session 21 capstone

What CareSlot does

- Reception registers patients (10-digit mobile, optional email and date of birth)
- Manages the doctor roster and availability
- Books visits: SCHEDULED → CHECKED IN → COMPLETED or CANCELLED
- Blocks double booking of a doctor or a patient, bookings in the past and edits to finished visits
- Live dashboard; works on phones



45
automated tests (pytest)

0
vulnerabilities in both images

53
AWS resources from Terraform

14
REST endpoints, 3 resources + stats

Tech stack



Frontend

React 19 · Vite 8 · Nginx (non-root)



Backend

FastAPI · SQLAlchemy 2 · Pydantic · Alembic



Database

PostgreSQL 16 · EBS gp3 volume



Testing

pytest - 45 tests · ruff lint



Containers

Docker · Compose · Alpine · multi-stage



CI/CD

GitHub Actions · GHCR · Trivy · gitleaks



Infrastructure

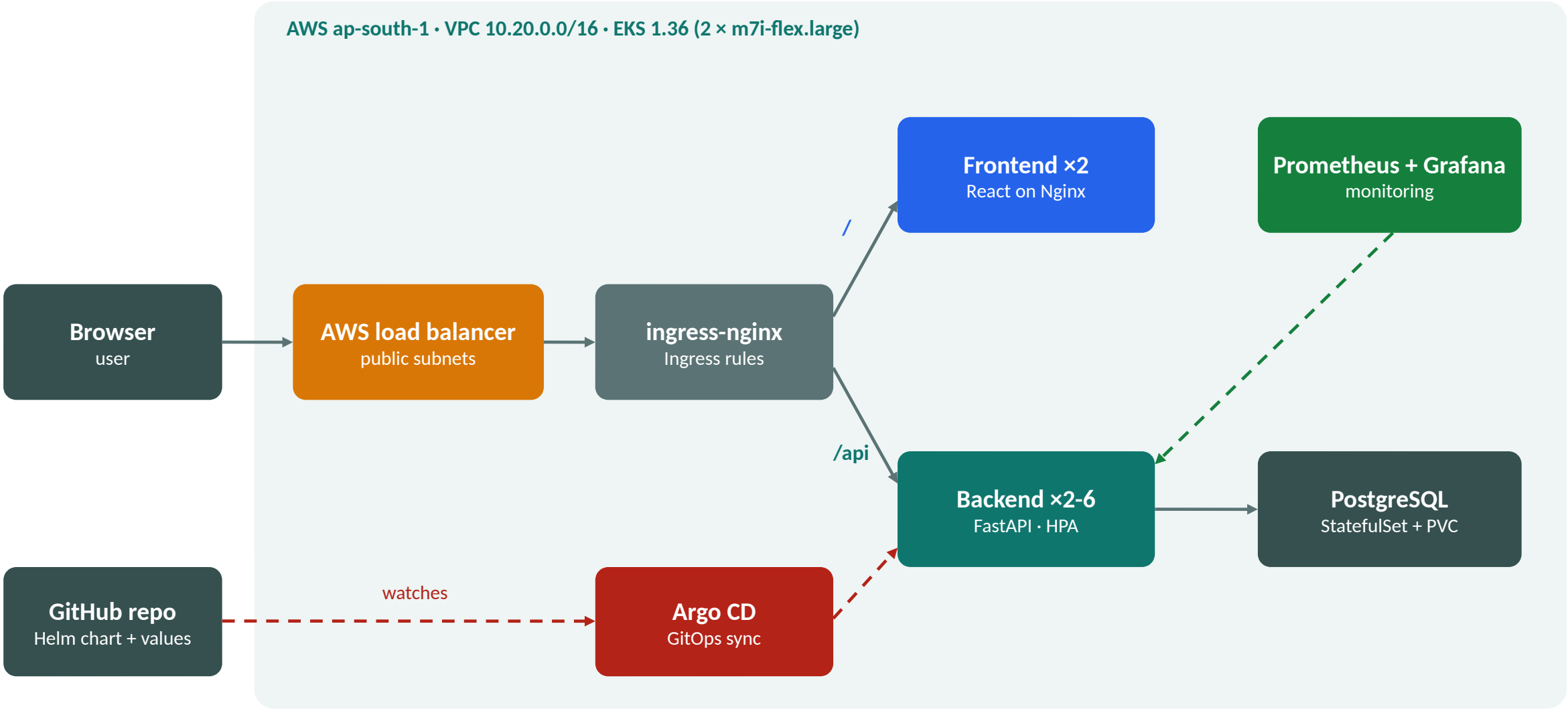
Terraform · AWS VPC · EKS 1.36



Kubernetes & ops

Helm · Ingress · HPA · Argo CD · Prometheus · Grafana

Architecture



CI/CD flow

1 **git push** to main

2 **Test** ruff · 45 tests · Alembic · React build

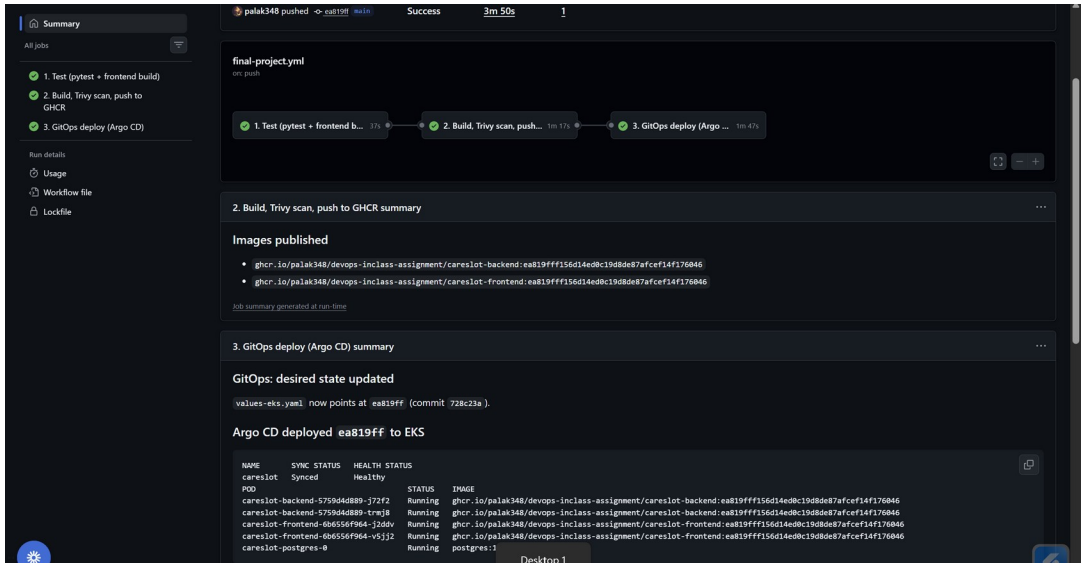
3 **Build** 2 Docker images

4 **Trivy gate** fail on any HIGH/CRITICAL

5 **GHCR** tag = commit SHA

6 **GitOps commit** SHA → values-eks.yaml

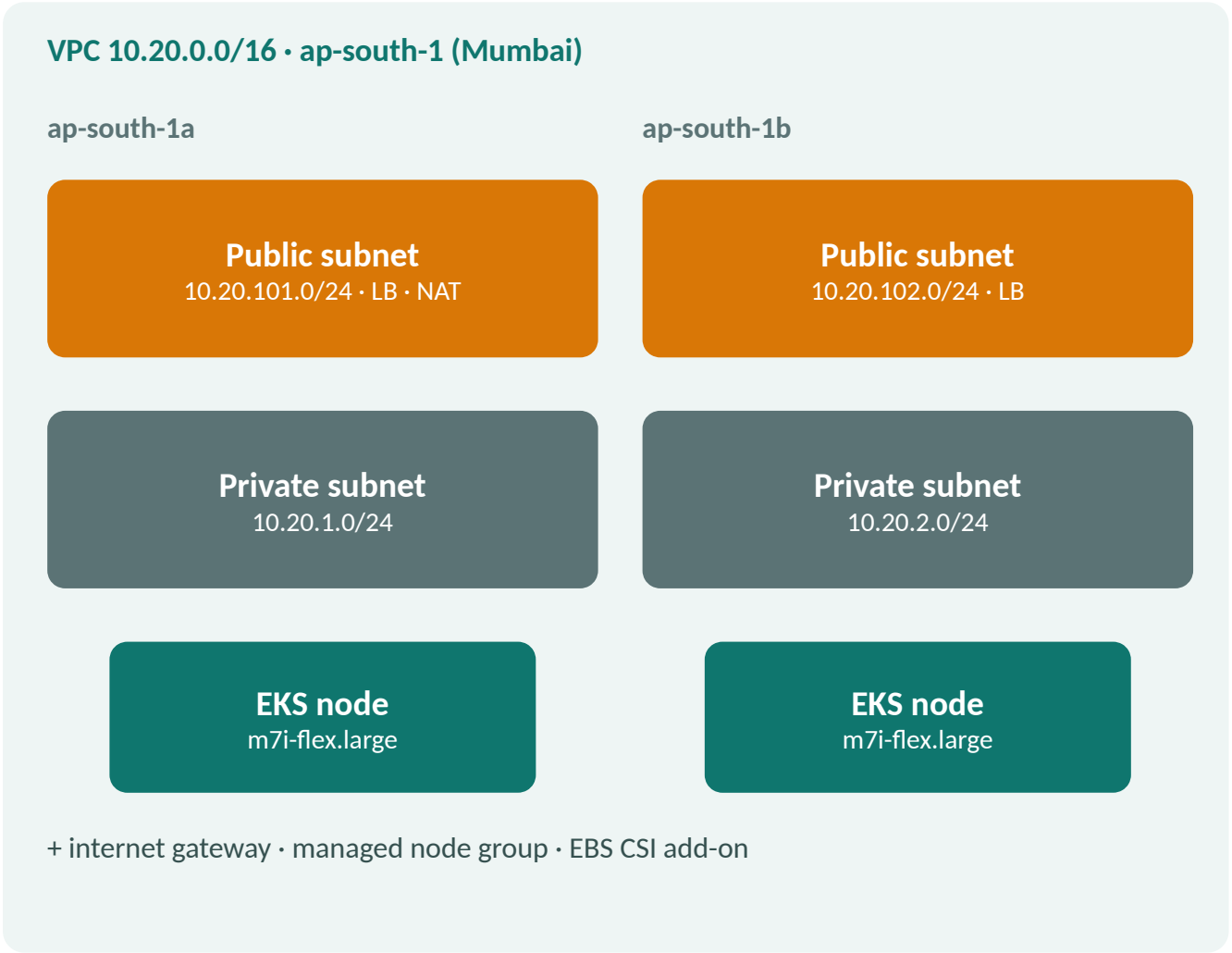
7 **Argo CD** syncs to EKS



What stops a bad change

- A failing test or lint error stops the run - no image is built
- A HIGH or CRITICAL CVE fails the Trivy gate - nothing is pushed
- Images are tagged with the commit SHA, never latest
- Deploy = a Git commit: Argo CD rolls it out, CI waits until it is Healthy

Infrastructure as code - Terraform



53
resources created by terraform apply

~$0.40
per hour, measured running cost

- Modules: terraform-aws-modules/vpc 6.7.3 + eks 21.29.0
- EKS 1.36 in standard support (extended support costs 6×)
- Free-plan account: only Free-Tier-eligible nodes - verified with a real launch
- No KMS key or log group, so terraform destroy leaves nothing billable
- No credentials in code; tfvars.example documents the variables

Kubernetes setup

Helm chart careslot

Deployments, ClusterIP Services, StatefulSet + PVC, probes

Ingress

/api → backend, / → frontend, behind the AWS load balancer

HPA

backend 2-6 Pods on CPU 60 %

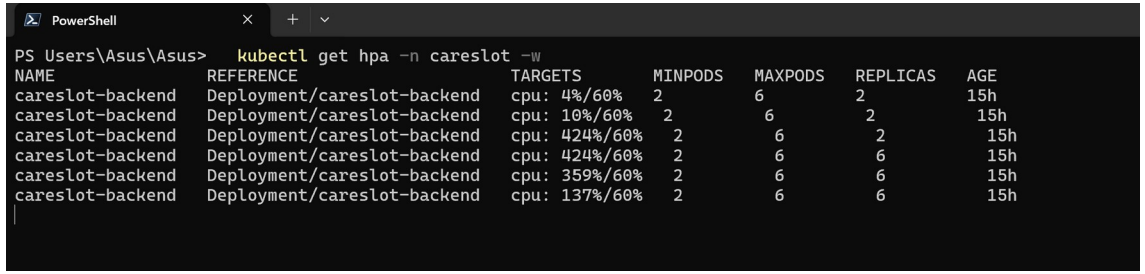
Argo CD (GitOps)

automated sync, prune and self-heal from Git

Hardening

non-root, capabilities dropped, Pod Security restricted

Autoscaling, measured



2 → 6

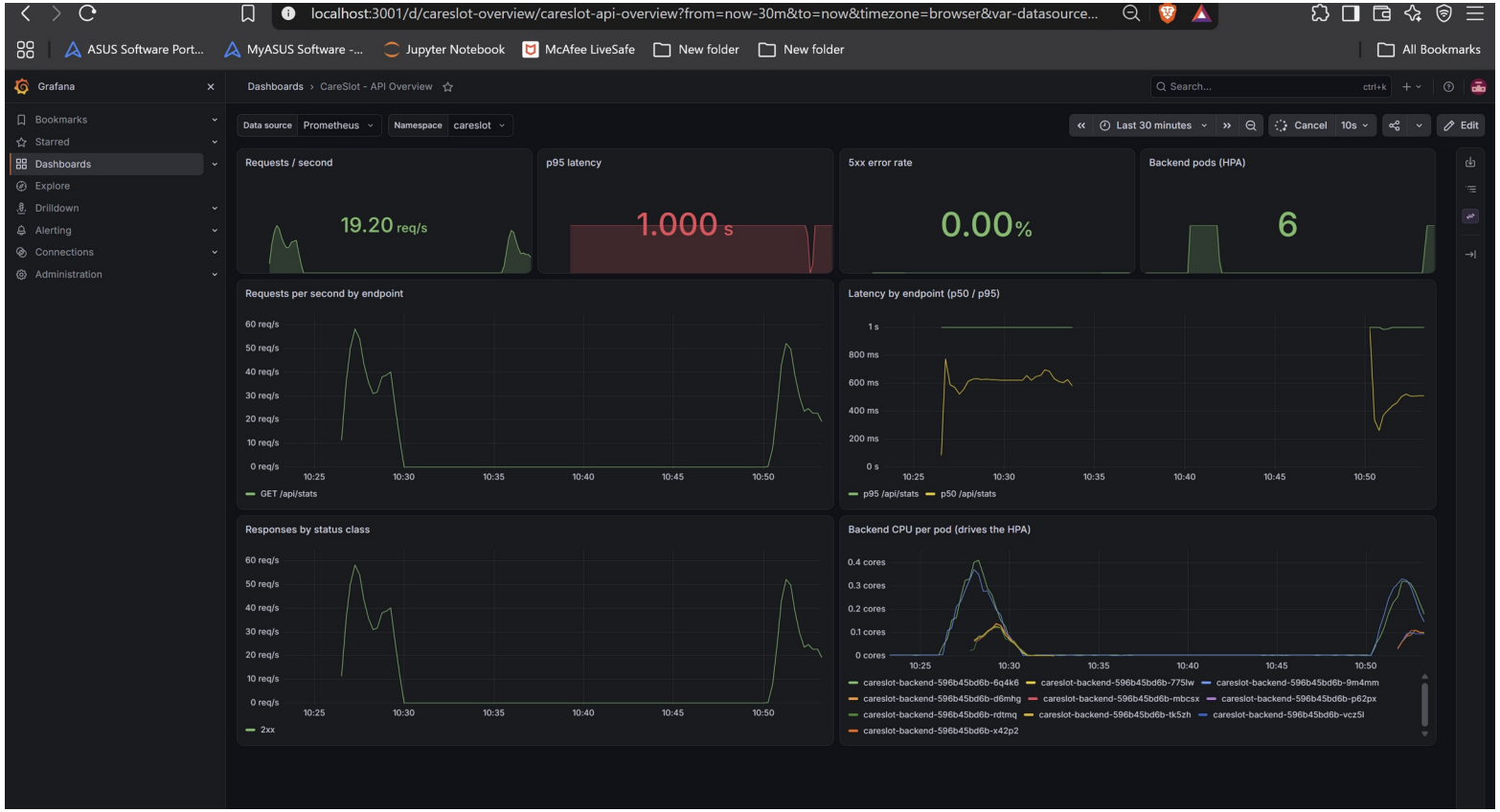
backend Pods when CPU reached 424 % of request

Synced

Argo CD Application · Healthy

Local: minikube + plain Helm · AWS: EKS + Argo CD

Observability



Under load: 19 req/s, 0.00 % errors, 6 backend Pods

- FastAPI exposes /metrics (Prometheus format)
- A ServiceMonitor makes Prometheus scrape every backend Pod - targets UP
- Grafana dashboard: requests/s, p95 latency, 5xx error rate, Pods, CPU
- Installed with kube-prometheus-stack; dashboard shipped in the Helm chart

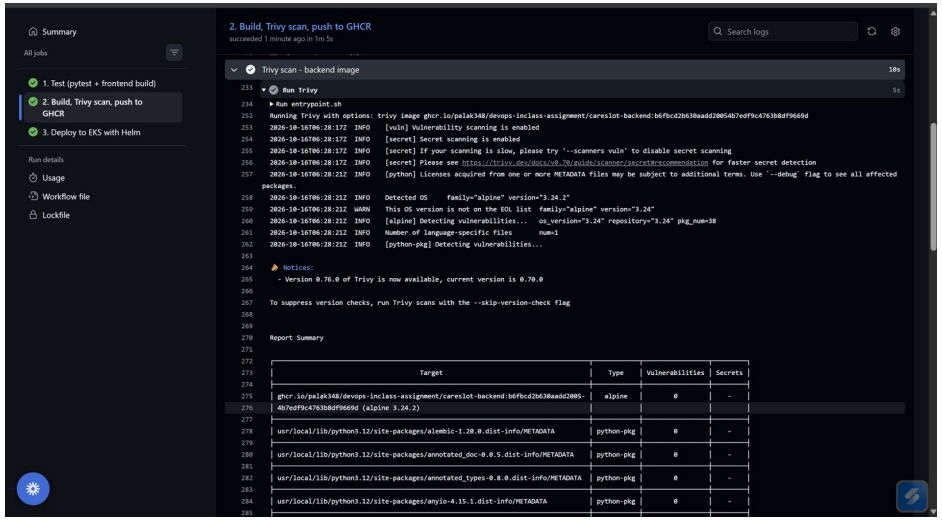
Security in the pipeline

44 → 0
HIGH CVEs in the backend image (Debian → Alpine, pip removed)

5 → 0
gitleaks findings in the full git history (reviewed demo strings)

3 / 3
containers run as non-root (uid 10001, 101, 70)

- Trivy scans both images before push and fails on any HIGH or CRITICAL CVE - even unfixed ones
- Example fixed: CVE-2026-85091, a heap overflow in zlib, removed by apk upgrade during the build
- No secrets in Git: AWS keys only in aws configure and GitHub secrets; DB and Grafana passwords generated in-cluster
- Read-only root filesystem for the backend, all Linux capabilities dropped, Pod Security restricted



CI run: Trivy Report Summary - alpine 3.24.2, 0 vulnerabilities

Key learnings

Verify the account, not the docs

A Free-plan AWS account can only launch Free-Tier types - t3.medium was blocked; a 1-minute real launch proved m7i-flex.large works.

Defaults can break storage

The EKS module's IMDS hop limit of 1 left the EBS CSI driver without credentials; hop limit 2 fixed the database volume.

GitOps changes chart design

Argo CD renders charts without Helm lookup, so the random DB password needed an existing Secret.

Don't hide CVEs, remove them

ignore-unfixed hid 44 HIGH CVEs; a smaller base image and no pip removed them.

Test in a real browser

A headless browser run found a date-picker bug that blocked bookings - unit tests had passed.

Plan the teardown

Load balancers created by Kubernetes block terraform destroy - uninstall them first.



Thank you

Next steps: login and roles · HTTPS with a domain · Amazon RDS · S3 Terraform state · alerting

github.com/palak348/devops-inclass-assignment